

BedSAT: Antarctica

Exploring what lies beneath using big data and modelling

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Aims

BedSAT: Antarctica is a modelling approach that combines satellite observations of the ice surface, airborne radar surveys of the bed, and numerical ice flow modelling. It is physically informed by observed bed morphometry statistics which feed into synthetic bedrock generation for ISSM ice-sheet models. With BedSAT we aim to use machine learning to model bed-to-surface transfer from physics-based simulations over these synthetic beds, and to invert the transfer process, so that observed surface expression from real satellite data can be mapped back to bed topography. The goal of this project is to improve the resolution and accuracy of Antarctic bed topography.

1.1 Project Deliverables

- **Paper 1 — ODSA:** Develop a physically informed spectral analysis framework for quantifying Antarctic subglacial landscape characteristics and classifying bed morphology into archetypes for ice sheet modelling;
- **Paper 2 — Bedrock factory:** Design a synthetic bed topography generator for Antarctic landscapes that takes morphometric statistics from ODSA as inputs and produces bed realisations at ice sheet model resolution.
- **Paper 1 — BedSAT:** Develop a modelling approach that combines satellite remote sensing, airborne radar and numerical ice flow modelling to improve the resolution and accuracy of Antarctic bed topography.

1.2 Timeline

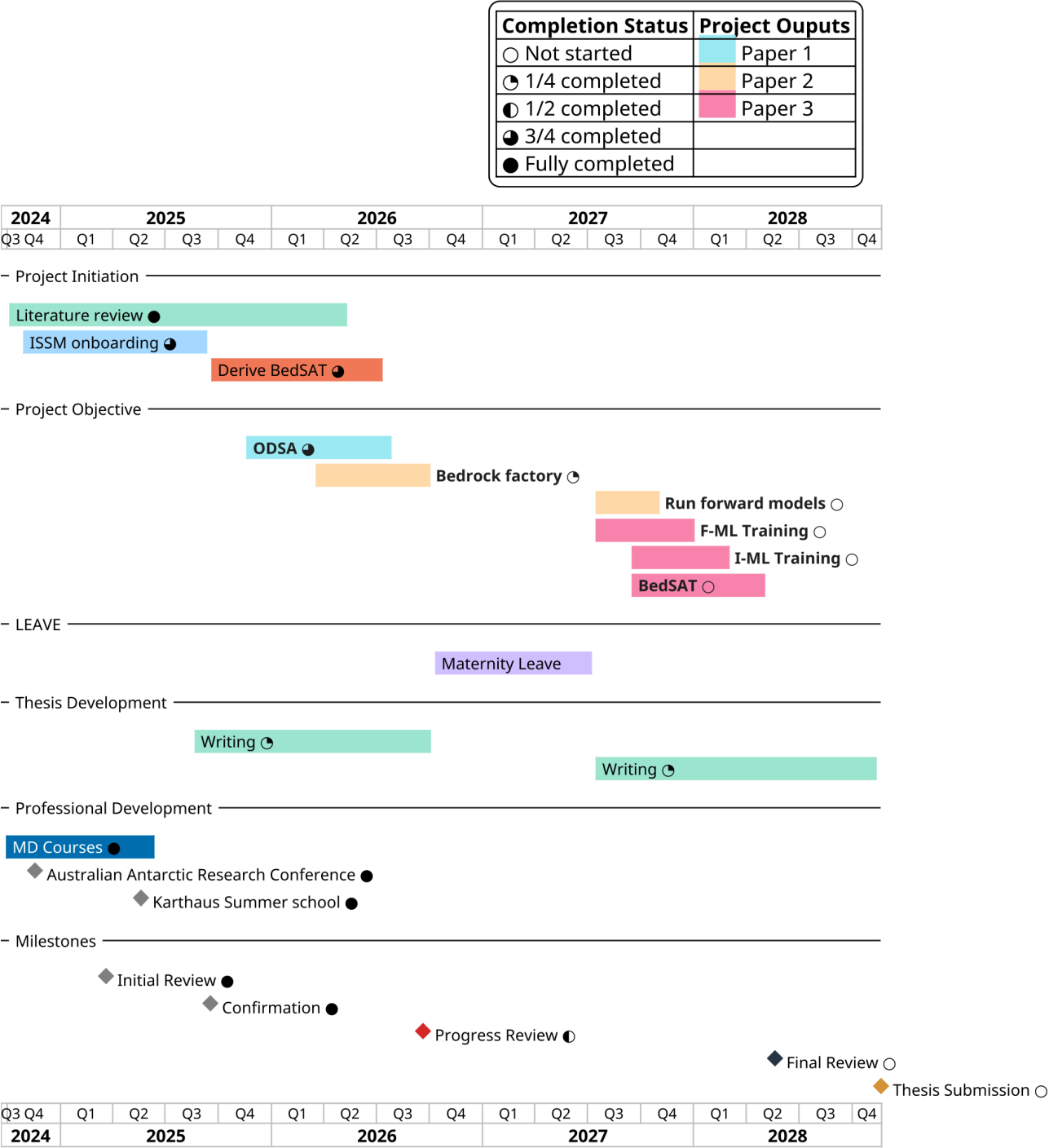


Figure 1.1: Project timeline. In the Project Objective section, the colour of each bar shows which paper the task contributes to. The circle next to each item is an estimate of how complete it is: a quarter-filled circle means that the prototype of the code and workflow exists. A half-filled circle means that the task workflow is in progress. A three-quarter-filled circle means that the results are complete and a paper needs to be written up or the task needs to be officially approved by the panel. A filled circle means that the task is complete.

Preliminary Literature review

Bed topography is one of the most crucial boundary conditions that influences ice flow and loss from the Antarctic Ice-Sheet (AIS) [1]. Bed topography datasets are typically generated from airborne radar echo sounding (RES) surveys, which are sparse and unevenly distributed across the Antarctic continent. Interpolation schemes used to “gap fill” these sparse datasets yield bed topography estimates that have uncertainties of multiple hundreds of metres in elevation [1]. Some examples of these products are the Bedmap ice bed, surface and thickness gridded datasets for Antarctica [2–4], or BedMachine [5,6]. These uncertainties then propagate through simulations of AIS evolution under climate change [7]. Given the logistical challenges of accessing large parts of Antarctica, there is a need for alternative approaches that integrate diverse and possibly more spatially complete data streams, including satellite data.

2.1 Approaches to Bed Topography Reconstruction

To address the gap in understanding subglacial conditions, the research goal of this project is to develop a method that combines forward and inverse modelling, leveraging high-resolution satellite surface data to significantly improve bed topography estimates in regions where direct radar measurements are sparse. The generation of gridded bed topographies in Antarctica is based on two primary modelling approaches: forward and inverse models.

Forward models aim to understand how different bed properties influence ice flow. These often generate interpolated gridded topography using geostatistical techniques like kriging [8] and are valuable for investigating how uncertainties in the bed can impact outcomes like simulated ice mass loss. Forward models can generally be classed as elevation-preserving or texture-preserving (the bed’s geostatistical properties are maintained). Their limitation is the reliance on assumptions about the bed conditions rather than direct observation, so they explore possibilities instead of determining the actual topography.

Inverse models involve methods that assimilate other data sources and work by inferring bed properties, such as topography or basal slipperiness, from surface observations like ice elevation and velocity. Techniques like control method inversion [9] and 4dvar assimilation minimise mismatches between observed and simulated data. On the other hand, mass conservation approaches use physical laws to reconstruct the bed, especially where measurements are sparse [1,5]. Despite their power, these methods face significant challenges. Variational approaches often require regularisation to prevent non-physical results and over-fitting [10], and time-dependent methods like 4dvar are computationally expensive. Probabilistic methods such as Markov Chain Monte Carlo (MCMC) quantify uncertainty well but remain too computationally intensive [10] for large-scale models.

Ultimately, the limitations of these methods reveal a core challenge in modern glaciology: the need to develop a new approach that can fully utilise rich satellite surface data to overcome the lack of direct subglacial measurements.

2.2 Quantifying Bed Morphometry

It is still uncommon for methods to systematically quantify the relationship between bed topography and its morphometric landscape statistics. Studies that measure the statistics report them as continuous fields, and terrain types are assigned to specific regions rather than to individual cells, with each parameter described qualitatively. Siegert et al. [11] define three bed roughness groups across named areas of East Antarctica, and Rippin et al. [12] assign three terrain types to the areas of their survey. Numerical thresholds have been published for single parameters, and Jordan et al. [13] divide the Hurst exponent at 0.5 and 0.75 to define low, medium and high roughness categories, but no scheme applies thresholds to a combination of axes per analysed cell. For published classified maps, each cell is often assigned to one class with no margin reported alongside it: Ockenden et al. [14] map the Antarctic bed into five landscape classes on 50 km cells, and their accompanying data release [15] records each assignment without a confidence interval or an alternative class. For classification schemes in general, ambiguity in landscape class is often only acknowledged as a caveat or analysed under a qualitative lens of multiple interpretations but it is not quantified, so a published class cannot be read alongside the alternatives that the measurement might still admit. That absence of margins makes such maps vulnerable to survey artefacts, like local roughness measurement bias due to radar migration status [11, 16, 17] or measurement limitations from the orientation and along-track sampling resolution due to uneven survey coverage [18].

Assigning a single class to a region or cell also averages the terrain within it. Rippin et al. [12] note that their large-scale grouping masks some complexities within the groups, and the selective erosion class of Ockenden et al. [14], which covers 56% of Antarctica, is defined as all topography that is not clearly “low-relief” or “alpine”. Such classes cannot capture the gradual change in roughness reported by Bingham and Siegert [19] or the continuum of landscape types described by Sugden [20].

In chapter 5, we outline the progress for Stage 1 of BedSAT: Observational Data Spectral Analysis (ODSA), a physically informed framework for quantifying Antarctic subglacial landscape characteristics and classifying bed morphology into archetypes for ice-sheet modelling. The landscape classification provides an empirical foundation for the generation of structurally realistic synthetic bed topographies. ODSA operates on the along-track picks published with Bedmap3 instead of a gridded product, which allows us to measure the survey data directly and avoid the interpolation issues described earlier. The tracks used are visible in Fig. 2.1.

2.3 From Signal Transfer to Tractable Inversion

This project aims to develop a framework to connect observable surface features to the subglacial bed below, ensuring that the governing physics of ice-sheet flow are respected. To do this, our work focuses on utilising surface-bed transfer functions.

The core principle behind a transfer function is that the ice-sheet acts as a physical filter, modifying the expression of bed features as they propagate to the surface. The work by Budd [21] established that ice preferentially dampens short-wavelength bed undulations while features with a wavelength approximately $3.3\times$ the ice thickness are most clearly expressed at the surface. This process also introduces a phase lag, and can be described mathematically using frequency-dependent transfer functions. More recent studies have built directly on this foundation. For instance, Gudmundsson and Raymond [22] refined the transfer function concept for ice streams, and Ockenden et al. [23] applied it in reverse, using full-Stokes transfer functions to invert high-resolution satellite observations of surface elevation and velocity to infer bed properties. These studies collectively establish that the physical relationship between

the bed and the surface provides a viable pathway for subglacial mapping.

Despite the success of these approaches, their practical application has been limited by simplifying assumptions about ice physics. A primary limitation is the reliance on a linear (Newtonian) ice rheology, where stress is directly proportional to the strain rate (i.e., a “constant viscosity”). This assumption is often utilised to make the inversion mathematically tractable, but it contrasts with the widely accepted non-linear Glen’s Flow Law, where the stress exponent is typically $n \approx 3$ or even $n = 4$. The choice of rheology is a critical control on the bed-to-surface signal transfer; a non-linear rheology ($n = 4$) produces significantly different surface expressions compared to a linear one ($n = 1$), and by largely ignoring non-linear rheology and complex basal sliding conditions, past inversion methods have introduced uncertainties and may not be robustly applicable across all dynamic regimes of the ice-sheet.

The opportunity lies in exploiting the vast wealth of underutilised high-resolution satellite surface observations including NASA’s MEaSUREs [24] velocities and REMA elevations [25]. Our approach with BedSAT will harness these data streams by building upon established transfer function theory while addressing the fundamental limitation that has restricted past approaches: the mathematical intractability of non-linear ice physics.

BedSAT will connect surface observations with bed topography using more realistic rheological assumptions ($n = 4$) and complex sliding conditions, rather than the simplified linear physics commonly used by traditional inversions. The key step to making this process tractable is Physics-Informed Machine Learning (Physics-ML), which solves the computational bottleneck that has forced previous methods to rely on unrealistic simplifications. By leveraging NVIDIA PhysicsNeMo, a framework designed to blend governing physics (PDEs) with training data [26], BedSAT can learn the non-linear mapping between surface expressions and bed topography without linearising the physics. This approach would transform what was previously computationally intractable into a fast, accurate inverse solver. Through an iterative process where initially inverted bed topography is integrated into forward models for progressive refinement, BedSAT will deliver physically consistent reconstructions validated against independent datasets. This would represent a major improvement: where previous methods had to choose between physical realism and computational feasibility, Physics-ML can enable both simultaneously.

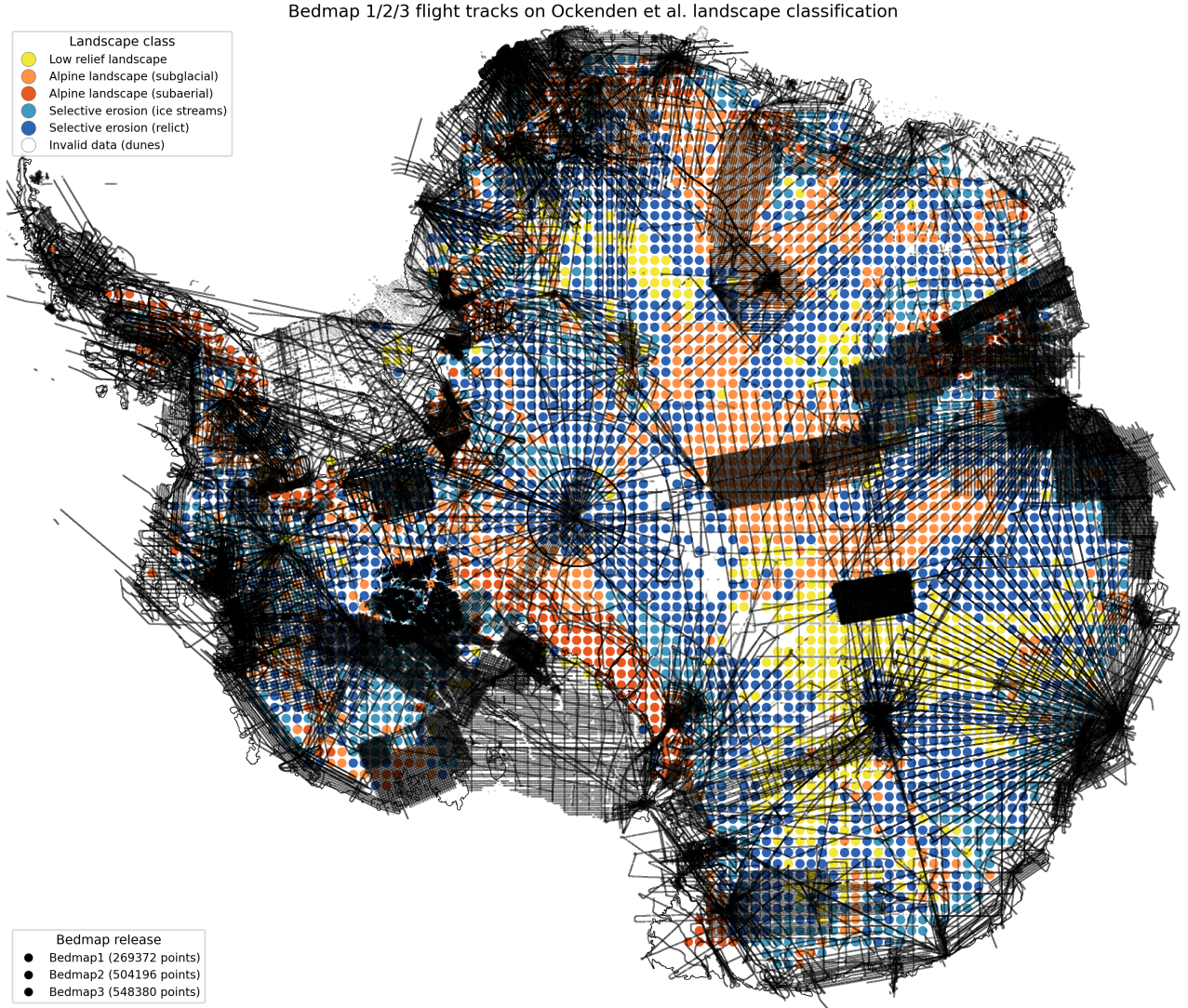


Figure 2.1: Bedmap1, Bedmap2 and Bedmap3 along-track survey points (black) over the continental landscape classification map by Ockenden et al. [14] (coloured cells, cells masked as invalid data (dunes) are drawn white). We reproduce the classification from the published metric grids [15] on 50 km cells. The point counts in the legend show the number of subsampled plotted points rather than published picks. ODSA operates on these along-track data, so the figure shows the sampling available to us across a continental landscape classification.

Research Significance

The accelerating loss of continental ice-sheets in both Greenland and Antarctica are major contributors to global sea level rise [27]. Impacts extend beyond direct ice loss: as fresh water from melting ice-sheets is added into the ocean, it increases ocean stratification disrupting global thermohaline circulation [28]. In addition, cold freshwater can dissolve larger amounts of CO_2 than regular ocean water creating corrosive conditions for marine life [27]. While there is high confidence in current ice loss and retreat observations in many areas, there is more uncertainty about the mechanisms driving these changes and their future progression [29]. Uncertainty increases in regions with variable bed conditions, where characteristics like bed slipperiness and roughness have been difficult to verify via direct observations. Other problematic areas involve the ice-sheet grounding line (GL): The zone that delineates where the ice-sheet is grounded on bedrock. The GL retreat rate depends crucially on topographical features like pinning points [29], which lead to increased retardation of upstream flow. Although this mechanism is established, major knowledge gaps persist in mapping bed topography across Antarctic ice-sheet margins, with over half of all margin areas having insufficient data within 5km of the grounding zone [30]. Addressing this data gap through systematic mapping enhanced by auxiliary data streams with more complete coverage would significantly improve both our understanding of current ice dynamics and the accuracy of ice-sheet models projecting future changes.

Subglacial bed roughness is already being used as a tool to infer basal processes and ice-sheet history [18], and it may provide information on basal drag, which is a key boundary condition for ice-sheet models [14]. In many regions of Antarctica the spacing between radar survey tracks is 10 to 100 km [14], but away from survey tracks, interpolated bed maps have not been able to reproduce the roughness of subglacial terrain observed along radar profiles [14]. A map like Bedmap3 with up to 93% of the 500 m ice-thickness grid cells filled with interpolated values [4] has an insufficient resolution to be used directly in ice-sheet models.

As part of the first stage of BedSAT, ODSA quantifies subglacial landscape characteristics directly from the radar profiles and classifies bed morphology into landscape archetypes for synthetic bedrock generation for ice-sheet modelling. Previous studies have characterised bed roughness with one or two spectral parameters [11–13, 18, 31, 32], but ODSA describes each analysed window of a RES profile with a landscape vector of measured elements (see tables 4.1 and 4.2). Four of these elements serve as classification axes (“classifiers”) and are compared with a literature informed catalogue of landscape archetypes. Each of the four classifiers carries an uncertainty envelope, and the output for a window is the set of archetypes that the measurement cannot rule out. The remaining vector elements, which we consider “descriptors” have no thresholds in the literature and are reported as continuous values. When an analysed window is reported as landscape degenerate the ODSA pipeline names the axis responsible for the overlap and reports the observable required to break the tie. These capabilities impact survey design by telling us which instrument to fly in a future survey. Additionally bed texture decorrelation between nearby transects constrains the track spacing a survey needs to resolve the landscape it is flown to characterise.

Project Methodology

4.1 Development of the BedSAT Inversion Framework

BedSAT's goal is to leverage a physically informed transfer function with improved understanding of how ice rheology and basal sliding conditions affect the surface expression of subglacial topography. Its development requires the following stages.

1. **Training:** BedSAT will be trained and validated using a physically informed (ODSA) synthetic bedrock database (Bedrock Factory) and a data-rich region, such as the Aurora Subglacial Basin, where extensive ice-penetrating radar data of the bed is available for direct comparison.
Forward Ice Flow Model: The forward model of ice dynamics will be configured for the chosen study site, including key model assumptions, such as the choice of ice flow equations and the parameters governing ice rheology. The robustness of the numerical configuration will be confirmed through grid independence testing to ensure that the results are not an artefact of the model's spatial resolution.
Inversion Pre-processing: The primary observational inputs for the model will be satellite-derived surface velocity, surface elevation [24, 25] and ice thickness [33, 34]. A critical pre-processing step will involve filtering high-frequency noise (e.g., aeolian features) from the surface data to ensure that the glaciological signal from the underlying bed is isolated.
2. **Inverse Modelling to Estimate Bed Topography:** The core of the BedSAT is the inversion. BedSAT will iteratively adjust an initial estimate of the bed topography, aiming to minimize the mismatch between the forward model's simulated surface velocities and the satellite-observed velocities.
3. **Validation and Sensitivity Analysis:** BedSAT's performance will be rigorously evaluated. First, the inverted bed topography will be directly validated against the "ground-truth" radar-derived bed maps for the study region. Second, a series of sensitivity analyses will be conducted to quantify the model's robustness. These tests will systematically alter key assumptions, such as the rheological parameters for different regions, to determine their impact on the final inverted topography and to establish the uncertainty bounds of the framework's results.

4.2 ODSA's Landscape Classification Vector

To inform the synthetic bedrock database that will be used in the training, we need bed morphometric statistics to describe a landscape class. We apply ODSA to seven reference regions as proof of concept (see Fig. 4.1), quantifying how well the flight lines sample their survey footprint.

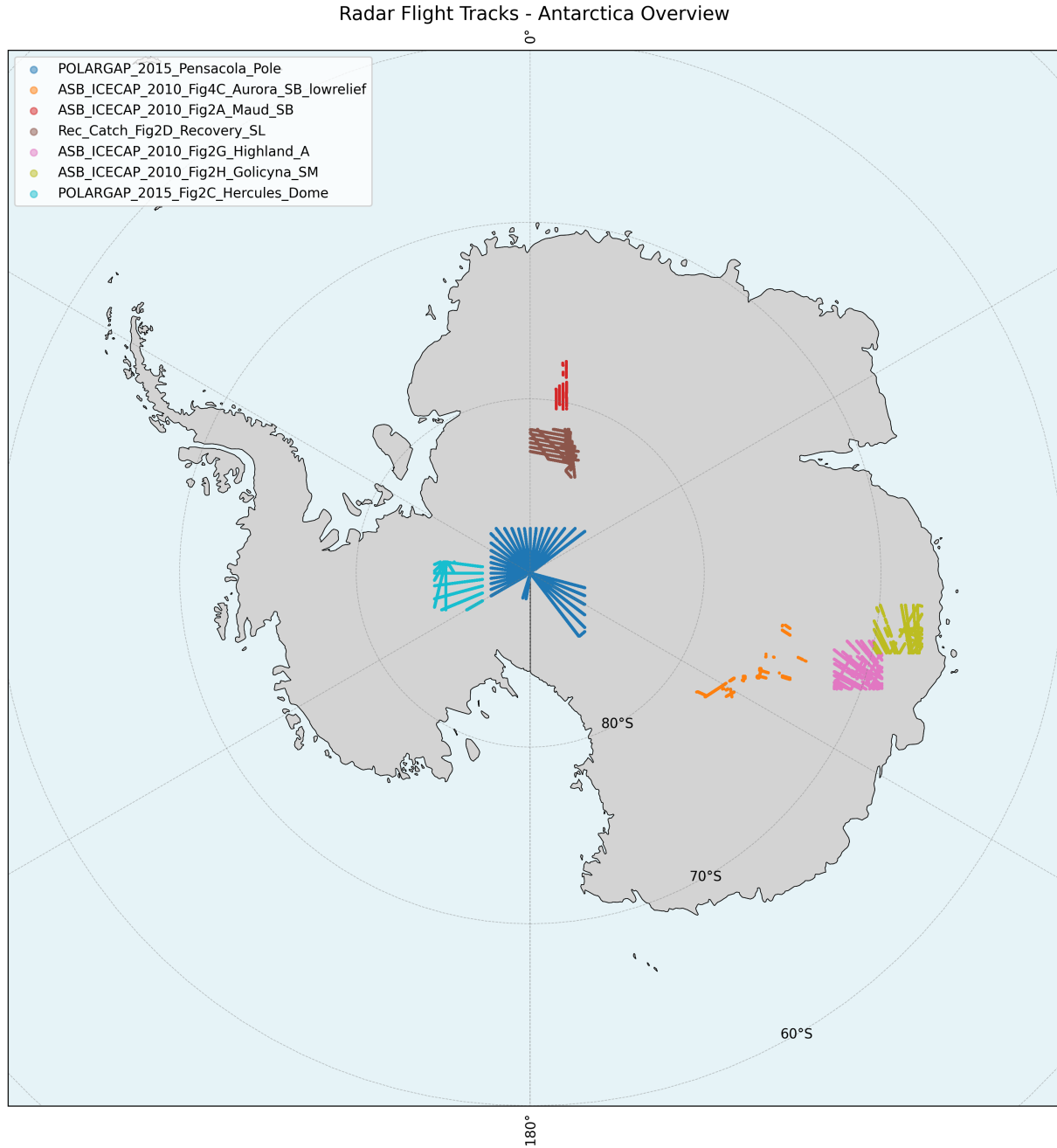


Figure 4.1: The seven RES survey regions we analysed with ODSA. Five regions use the PS71 boxes from [15]. The Aurora low-relief region is restricted to the low-relief cells of the Ockenden classification, and the Pensacola Pole Basin region uses a box that we defined.

Our proposal is that to bound subglacial landscape class it is necessary to have multiple measured statistics that characterise different landscape properties. ODSA’s landscape vector carries 11 elements measured per window. Four classify and seven do not. Each classifier carries an uncertainty envelope of two standard deviations at either side of the measured value, taken from the spectral fit covariance in the case of the roughness slope (β) and from measurement error budgets for relief, velocity and elevation. The landscape vector provides the morphometric statistics that the Bedrock Factory suite will take as inputs to generate synthetic beds at ice-sheet model resolution.

Table 4.1: Classification axes of the landscape classification vector. Across all four, the framework classifies each axis on value $\pm 2\sigma$, so a window can be admissible in more than one band

Variable	Description/Thresholds
Classifiers β	Slope of the bed power spectrum over the fitted band. A measure of bed roughness shape: low β distributes power to the short wavelengths; high β concentrates it at long wavelengths. The fit band spans 250 m to 50 km. The class thresholds are: chaotic < 1.5 , hard 1.5 to 2.0, transitional 2.0 to 2.5 and soft ≥ 2.5 . The 2.0 and 2.5 breaks follow the Hurst-exponent splits of [13]. The 1.5 break is an operational choice. Beds in the hard class are interpreted as scoured crystalline terrain, like in the areal-scouring landscapes of [20].
relief_m	Elevation range within a window, calculated as $\max(\text{bed}) - \min(\text{bed})$ over the raw 50 km profile, in metres. It gives the amplitude of the largest structure a window contains. Regional slope enters the range directly. Relief class thresholds are set to flat: < 350 m; mountainous: ≥ 800 m; while values in between these are labelled as subdued. these are set based on the median relief for the tested regions.
measures_speed_mean	Mean MEaSUREs InSAR surface speed at the window’s track points, sampled over the same points that enter the spectral fit. Bands are very_low (< 5), low (5 to < 10), moderate (10 to < 50) and fast (≥ 50) all in m/yr and set operationally. Antarctic surface speed is bimodal, with a slow mode at $\approx 5 \text{ mm/yr}$ and a fast mode at 250 m/yr, and measured speed departs from internal deformation above about 15 m/yr [24]. Each window carries a sampled error from the ERRX/ERRY fields of that product.
bed_elev_mean	The mean bed elevation within a window, in metres above present-day sea level. Class thresholds are set to submerged: < 0 m; elevated at ≥ 1000 m, and emerged for values in between. Motivated by [11, 12, 35].

Table 4.2: Descriptors of the landscape classification vector, these are continuous variables with no breaks.

Variable	Description
A_1km	$\log_{10}$ of bed power spectrum at a 1 km reference wavelength. Evaluated from the fit as $\text{psd_intercept} + 3\beta$. It gives magnitude of the roughness in $\log_{10}(\text{m}^3)$ and fixes the level inside the fitted wavelength band. It carries a propagated uncertainty that includes the slope and intercept covariance.
rms_roughness	Root mean square height [?] of the window's detrended elevation residuals, in metres. It gives the amplitude of the topography with the regional ramp removed, which distinguishes it from relief. It depends on profile length, and is retained for comparability with the space-domain roughness literature rather than as a separate degree of freedom.
ξ	Integral of the bed power spectrum over the bedform band spanned from 300 m to 1200 m, in m^2 , it is the numerator of η [32]. Supported by [36, 37].
η	Characteristic wavelength for the window, in metres, following [32]. It square root of the weighted harmonic mean of the square of the wavelength at each frequency in the band spanned from 300 m to 1200 m with the spectrum as the weight.
hill_count	Number of distinct local maxima in the window's detrended bed profile, similar to the bumpiness metric by [14] but in transect form (1D). A point counts if it is the maximum of a 5 km box centred on it and the box relief exceeds 20 m; adjacent flagged points count once and maxima touching a window edge are dropped. Two hills must be at least 2.5 km apart, so the count cannot exceed 10 per 50 km window.
skewness	Third standardised moment of the window's detrended, untapered bed elevation residuals (bias-corrected). Positive values imply narrow highs on a broad low base, negative values mean narrow lows cut into a broad high surface, following the qualitative hills-against-troughs distinction in [20]. It is the only phase-asymmetry information in the vector.
kurtosis	Excess kurtosis of the window's detrended, untapered bed elevation residuals (bias-corrected). Positive values mean a few extreme highs or lows against a flat background, negative values mean a flat-topped or evenly spread distribution.

Progress

The ODSA pipeline feeds on the REMA surface elevation model [25] ice thickness and the local ice-flow direction. The flow direction comes from the REMA surface gradient with a baseline of $10\times$ the ice thickness following the smoothing function recommendation from McCormack et al. [?]. ODSA also uses the 450 m MEaSUREs v2 product [24]. At each point it gives the speed and the angle between the MEaSUREs velocity direction and the REMA flow direction. Allowing us to obtain the velocity uncertainties per window. A 50 km sliding window at 50% overlap is the unit in the spectral analysis of the bed RES data, (Not every window is 50 km. A segment shorter than 50 km is analysed as a single window the length of the segment).

The detrended bed profile for each Hann tapered window is analysed with a Lomb–Scargle periodogram (over 500 geospace frequencies), which accommodates uneven along-track sample spacing [?].

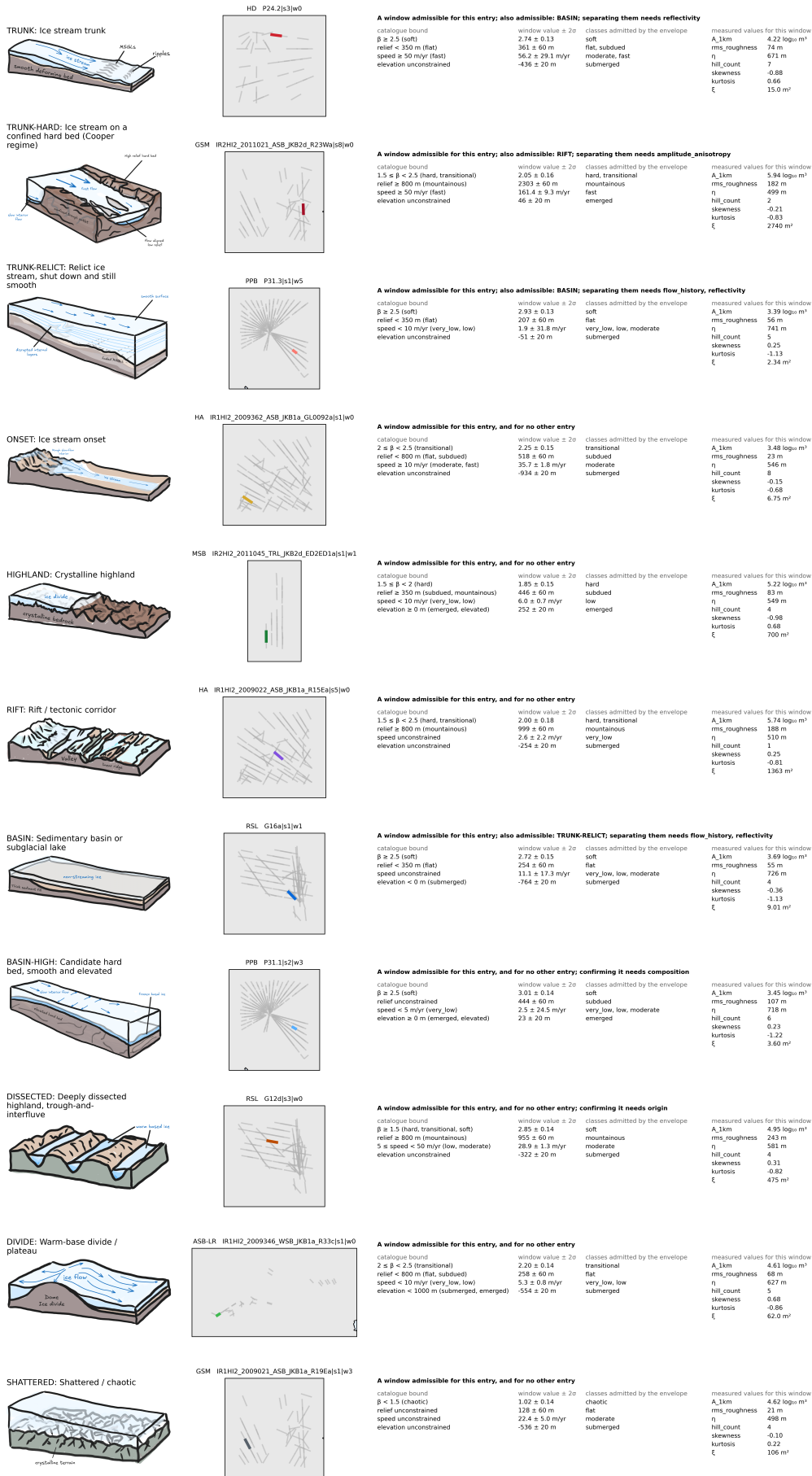


Figure 5.1

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